

Assignment 1: Data Collection and Preprocessing for Foundation Model Pre-Training

Report

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1. Dataset Sources and Total Size

A multi-domain corpus was assembled from three public sources to satisfy the diversity requirement (Section 3): Wikipedia (encyclopedic), CC-News (news), and OpenWebText (general web text). All three were streamed directly from the Hugging Face Hub (streaming=True), so no full corpus was ever downloaded in bulk -- only the volume actually needed was pulled, which also directly satisfies the Section 8 optional extension “integrate streaming datasets to handle very large corpora.”

Domain / Source	Collected volume
Wikipedia -- wikimedia/wikipedia, 20231101.en	174,374 docs / 773.1 MB
News -- vblagoje/cc_news	144,080 docs / 322.1 MB
Web -- Skylion007/openwebtext	38,717 docs / 193.3 MB
TOTAL	357,171 docs / 1.288 GB

CC-News was chosen as the “Common Crawl subset” source (Section 3 lists this as an example category): it is a well-known, cleanly-labeled Common-Crawl-derived subset curated specifically for news content, mapping directly onto the required news domain without needing to process raw Common Crawl WARC archives from scratch.

2. Cleaning Strategies and Reasoning

The cleaning pipeline (scripts/02_cleaning.py, pure standard library) applies, in order: HTML tag stripping, markdown-link/header/emphasis normalization, citation-marker removal, lowercasing and whitespace collapsing, short-document filtering (<50 words), and exact-duplicate removal via SHA-256 hashing of the normalized text -- hashing the normalized rather than raw text so that formatting-only differences still count as duplicates.

Domain	Kept / Short / Duplicate
Wikipedia	165,380 / 8,994 / 0
News	112,970 / 17,006 / 14,104
Web	38,714 / 1 / 2
TOTAL KEPT	317,064 documents (88.8% of raw)

- News had by far the highest duplicate rate (14,104 docs, 9.8%) -- consistent with wire-service stories being syndicated near-verbatim across outlets.
- Wikipedia had zero exact duplicates (articles are unique by construction), confirming the dedup logic isn't over-firing - it reflects the source, not a pipeline artifact.
- OpenWebText passed through almost untouched (99.99% kept), consistent with it already being pre-filtered by its original curators.

3. Tokenization Choices

Parameter	Value
Tokenizer	GPT-2 byte-level BPE (AutoTokenizer)
Vocabulary size	50,257 (confirmed at runtime)
Block size / padding	512 tokens; final chunk padded with EOS (GPT-2 has no native [PAD])
Blocks produced	656,119 total -- Wikipedia 398,928 · News 165,254 · Web 91,937

Blocks/document differs sharply by domain (Wikipedia 2.41, Web 2.37, News 1.46) and Section 7's word-length analysis explains why: Wikipedia's long tail (max 34,909 words; 10.7% of docs over 1,500 words) drives its block count up disproportionately, while News is concentrated in the 100-699 word range (73.9% of docs) and mostly fits in one block.

4. Data Loader Implementation Details

Two implementations in scripts/04_dataloader.py, both verified working end-to-end:

- TokenBlockDataset (map-style, torch.utils.data.Dataset): loads the full tensor into memory, supports shuffle=True. Verified producing correctly-shaped (8, 512) batches from all 656,119 blocks.
- StreamingTokenBlockDataset (IterableDataset): reads cleaned.jsonl line-by-line, tokenizes on the fly, never materializes the full dataset in memory; shards lines across workers via line_idx % num_workers. Verified: with num_workers=2, two independent tokenizer-load log lines confirmed two real worker processes actually sharding the file.

Both share a collate_fn stacking input_ids/attention_mask/labels (labels = input_ids; causal-LM shifting happens inside the loss computation), satisfying Section 4.3's streaming-for-memory-efficiency requirement while keeping a simpler map-style path available.

5. Challenges Encountered

- Memory: the map-style Dataset loads the entire tokenized tensor (5GB+ at this scale) into RAM at once. The streaming IterableDataset exists specifically to avoid this bottleneck for corpora too large to fit in memory, per Section 4.3's explicit requirement -- at our current 1.3GB scale both fit comfortably, but the streaming path is what makes the pipeline scale to much larger corpora without a rewrite.
- Batching: GPT-2 has no native pad token, so batches of variable-length chunks can't be stacked without a placeholder. EOS was reused as pad, which only works safely because attention_mask is computed and passed alongside every batch so the model/loss never attends to or is penalized on padded positions.
- Dataset-loading API changes: Hugging Face datasets dropped Python-script-based loading entirely. The bare identifier "cc_news" failed with "Dataset scripts are no longer supported" and was replaced with the namespaced, Parquet-format "vblagoje/cc_news"; "Skylion007/openwebtext" had separately migrated to Parquet upstream and worked unmodified.
- Chunking bug caught by unit testing: an early chunk_tokens() dropped an entire document if it was shorter than block_size/4, instead of only dropping small leftover TAIL chunks from long documents -- which would have silently deleted every short document from the dataset with no error. Fixed to only apply the minimum-size rule to non-first chunks.
- Missing output directory: tokenization crashed on its last line (sample_data/ didn't exist) only after several minutes tokenizing all 317,064 documents. The full tensor had already saved, so recovery was a 2-line fix (create the folder, slice+save the sample from the already-computed tensor) rather than a full re-run; the script now auto-creates parent directories defensively.
- Benign tokenizer noise: GPT-2 logs "Token indices sequence length ... (N > 1024)" for every document over 1024 tokens, assuming direct model input. We chunk afterward instead, so no tokens are lost; silenced by raising tokenizer.model_max_length.

6. Reflections on Preprocessing Impact on Pretraining Quality

- Domain diversity changes the cleaning problem, not just the content: News' 9.8% duplicate rate vs. Wikipedia's 0% shows a pipeline tuned on one domain would badly under- or over-dedup another -- validating diversity during pipeline design, not just data collection.
- Deduplication at scale is not cosmetic: 40,107 documents (11.2% of raw) were removed before tokenization; left in, repeated phrasing would be seen disproportionately often, a documented contributor to verbatim memorization.
- Chunking strategy interacts with domain composition: since Wikipedia and Web average over 2 blocks/doc versus News' 1.46, non-overlapping chunking discards more inter-chunk context for the longer-form domains -- the implemented --stride argument would recover this at the cost of redundant tokens.
- A silent bug is more dangerous than a loud one: the chunking bug would have produced a smaller, subtly-biased dataset with no crash or warning -- reinforcing that preprocessing logic needs unit tests against synthetic inputs, not just an end-to-end "did it crash" check.

7. Data Quality Analysis (Optional Extension, Section 8)

scripts/05_optional_analysis.py computes domain coverage and word-length statistics directly from cleaned.jsonl (no tokenizer needed):

Domain	Docs	% corpus	Avg words	Min	Max
Wikipedia	165,380	52.2%	737.9	50	34,909
News	112,970	35.6%	423.1	50	17,455

Web	38,714	12.2%	829.7	81	18,903
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Length-bucket distribution (% of each domain's documents): News skews short and concentrated (10.5% under 100 words, 73.9% between 100-699); Web skews longest (37.2% over 700 words, 11.2% over 1,500); Wikipedia is bimodal -- 9.1% are near-stub articles under 100 words while 10.7% exceed 1,500, the widest spread of the three domains.

8. Tools and Libraries Used

- Python 3.12
- Hugging Face datasets (streaming dataset loading) and transformers (AutoTokenizer, GPT-2 BPE)
- PyTorch (torch.utils.data.Dataset / IterableDataset / DataLoader), run on Apple Silicon (MPS backend)
- Python standard library only for cleaning and analysis: re, hashlib, json, collections